Zuse: A Decision Model for European Languages

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Abstract

Decision models answer fixed-choice questions about a document with a probability for each option and generate no text. Published evaluations of such models report little about languages other than English. We present Zuse, a family of three decision models built by adding a low-rank adapter and a pointer head to frozen open-weight language models, trained for 27 European languages. On 20,044 rows from European sources excluded from training, which also served for model selection and should be read as development results, the largest model reaches 87.5% accuracy with a pooled expected calibration error of 0.016 and a row-weighted per-source error of 0.035. Calibration is weaker within individual task groups. On English public suites the model is behind the closed Jev system on most reported measures, including a 14-dataset breadth index (47.5 against 53.3, difference 5.8, 95% CI 2.2 to 9.0). We also report selected findings from single-seed development runs. On an early corpus, a larger, newer base scored 10 points above a smaller one, though size and base quality cannot be separated. An added topic scheme cost about 10 points on a topic test in one isolated pair of runs, probably through label-word collision, and an injection augmentation that named only wrong options taught a shortcut rather than robustness. We report these results with the confounds and selection effects that limit them. Zuse can be tried at `https://bineric.com/zuse`.

1 Introduction

Many operational uses of language models require a choice among known options rather than generated text. Ticket routing, policy checks, moderation and form extraction all have this shape. A decision model suits such tasks: it reads a document and a set of questions, then returns a probability for each option. Because it cannot produce text, its output stays inside the options supplied, and its probabilities can be used to decide which cases to automate and which to send to a person.

This model class has developed quickly, with Jev [1], the open reproduction Kev [2] and Clef [3] all appearing in 2026. We found no per-language evaluation for any of them. Language coverage matters in Europe, where public bodies and companies work across many languages, including several with few speakers and limited training data. Whether the design transfers across such languages has not been measured in public.

This paper studies those questions through Zuse, a family of three decision models trained for 27 European languages. We make four contributions:

1. **A model family with detailed results.** We describe three models and report results per language, per task group and per evaluation suite, including suites where they fall behind other systems.

2. **Observations on training choices.** We report selected observations from single-seed runs and state known confounds.

3. **A language-reach measurement.** We test 210 languages on reading and topic classification, most of them absent from training.

4. **A measurement of automation limits.** We report how much of each task can be automated at a fixed error budget, and how this depends on chance level.

Section 2 reviews related work. Sections 3 and 4 describe the model and data. Section 5 sets out the experimental design, and Section 6 reports results. Section 7 analyses training choices, Section 8 discusses limitations and Section 9 concludes.

2 Related Work

Decision models. TypeSafe's Jev [1] introduced the typed request format of a state and named questions, trained with a method the company calls Reinforcement Learning for Calibrated Decisions. Its architecture, data and parameter count are undisclosed. Kev [2] is an open-weight reproduction on Qwen bases, and its public evaluation suites and serving code underpin much of the work reported here. Kev pairs a pretrained language model with a pointer head. Its three smaller models train a LoRA adapter on a frozen base, while Kev-27B fine-tunes every weight of a post-trained Qwen release. Clef [3] offers Jev-compatible open-weight models on the same Qwen bases. Zuse uses the adapter design at all three sizes and targets European languages.

Calibration and selective prediction. Temperature scaling [5] is the standard post-hoc method for calibrating neural classifiers, and the Brier score [6] and expected calibration error [7] are the usual measures. Reject-option classification goes back to Chow [8], and Geifman and El-Yaniv [9] applied risk-coverage analysis to deep networks. Our automation measure is an in-sample risk-coverage point, and we report its limits in Section 5.2.

Multilingual evaluation. Belebele [11] and SIB-200 [12] test reading comprehension and topic classification across many languages. ScandEval [15] provides Scandinavian benchmarks, including the acceptability data we use. We use these sets to measure transfer, and we add a comparison of language-model bases, including European-developed ones.

Prompt injection. Text inside a document can steer a language model [10]. Decision models are exposed to the same risk because documents are untrusted input, yet their fixed outputs make the effect simple to measure as a change in the selected option. We use that measure in Section 7.3.

3 Model

3.1 Architecture

Zuse uses a frozen open-weight language model as a reader (Figure 1). Each question forms one token sequence that holds the state, the instruction and the options, following the layout used by Kev [2]. Caller text is escaped so that it cannot contain the structural tokens. A pointer head reads the hidden states at the decision position and at each option, and gives each of the K options a score z_k. Probabilities follow from a single temperature T [5]:

$$p_k = \frac{\exp(z_k/T)}{\sum_{j=1}^{K} \exp(z_j/T)} \tag{1}$$

Yes/no questions are two-option choices. Score questions use ordered options, and the expected position gives the score.

In most evaluations, each row carries one question. In serving, the state is encoded once and cached so that several questions can share it. This reuses Kev's serving stack [2], which we did not modify in design.

3.2 Model family

Only a low-rank LoRA adapter [4] and the head are trained, with the base weights frozen. These two parts turn a text-generating language model into a decision model: the adapter changes how the base reads a request, and the head replaces text output with a probability over the supplied options. Table 1 lists the family. All three bases are published under Apache-2.0 [17].

3.3 Training

The adapter and head are trained with a cross-entropy objective and a permutation-consistency term, which penalises disagreement between a row and a copy of it with shuffled options. Augmentations include none-of-the-above and injection. In none-of-the-above augmentation, that option appears as both the correct and an incorrect answer, following Kev [2]. In injection augmentation, a planted instruction in the state names an option while the label stays unchanged, and that named option is sometimes correct (Section 7.3). The temperature is fitted on a development set by minimising negative log-likelihood. We do not disclose loss weights, schedules or other training settings.

4 Training Data

We do not disclose the sources, composition or size of the training data, which we treat as commercially sensitive. The models target 27 European languages. No customer data was used in training, and no output of a commercial API model was used. Section 8 sets out what this non-disclosure means for readers.

5 Experimental Setup

5.1 Evaluation suites

Table 2 lists the suites and how each was used. The status column matters for interpretation, because several suites influenced development.

The European suite contains Belebele reading [11], SIB-200 topics [12], multilingual MMLU [26, 13], two Norwegian knowledge sets [22], a sentiment set [25], Swiss court judgments [16], MInDS-14 banking intents [14], ScaLA acceptability in four Nordic languages [15], a German retrieval set [24] and French yes/no questions [23]. None of its sources appears in training, but some of its tasks resemble tasks seen in training. The suite therefore tests new sources rather than wholly new tasks.

Two sources in Transfer v4 need caution, because data used in training or in earlier runs may share origin with them. We did not confirm row-level overlap. Without those two sources, Zuse-27B's accuracy changes by about one point or less on both splits.

5.2 Metrics

For N decisions with confidence c_i, the highest option probability, we report accuracy, the multi-class Brier score [6] and expected calibration error over 15 equal-width bins [7].

The **automatable share** at error budget b is the largest fraction of decisions, taken in descending order of confidence, whose error rate stays within b:

$$A(b) = \max \left\{ \frac{n}{N} : \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}[\hat{y}_{(i)} \neq y_{(i)}] \leq b \right\} \tag{2}$$

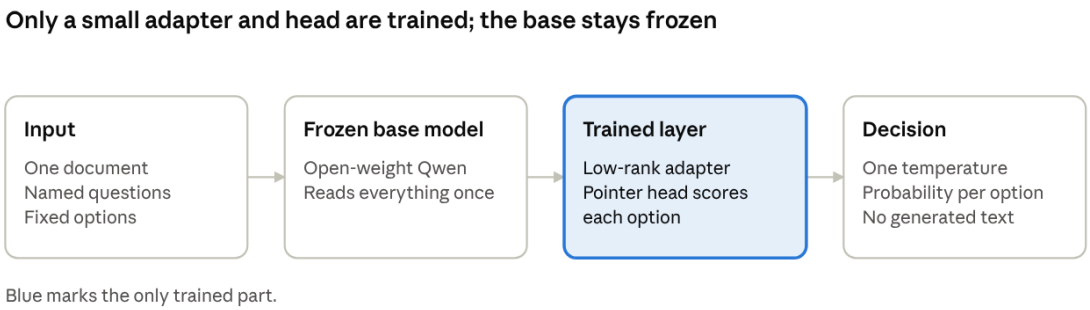


Figure 1: Zuse architecture: one pass from document to decision.

Table 1: The Zuse model family. Memory is resident memory when served.

Model	Base	Memory, 16-bit	Weights, 4-bit
Zuse-27B	Qwen3.8-27B	65.5 GB	15.1 GB
Zuse-9B	Qwen3.5-9B-Base	21.9 GB	5.6 GB
Zuse-4B	Qwen3.5-4B-Base	14.3 GB	3.1 GB

This follows the risk-coverage view of selective prediction [9]. The threshold is chosen on the rows it is scored on, and per-group figures use a separate threshold for each group, so $A(b)$ is an in-sample upper bound. Thresholds do not transfer cleanly: a threshold set on the Transfer v4 test split gives 8.9% error against a 5% budget on the development split, and in the reverse direction 2.4% error at 61% coverage. We write auto@5 for $A(0.05)$.

Order flips are the share of choice rows whose answer changes when options are reordered. We report this against reversed order and, separately, against a third random order, with and without averaging two orders. Both are measured on 1,223 choice rows from a 1,500-row sample of the European suite.

Planted-instruction following is the share of the same 1,500 rows on which the model selects an option named by a planted instruction. Its baseline is how often the clean model picks that option unprompted. The planted text uses one wording each in 10 languages and English for the rest, and always names a wrong option.

5.3 Selection procedure and statistical treatment

Almost all configurations were trained once. We selected models using the European suite together with the Transfer v4 development split, and used both to diagnose errors and to plan later runs. The English test splits were scored on five 27B checkpoints. Their accuracies span 0.859 to 0.866, and test scores were recorded alongside flagship decisions, so independence from selection cannot be claimed.

On a 764-row split, an accuracy near 0.86 has a 95% margin of about 2.5 points. For the European suite the margin is about 0.5 points under an independence assumption, and wider in practice because rows cluster within sources. Retraining on identical data moved two small-model results by up to 0.5 points, which is our only direct measure of training noise. Seeds for those retrainings are unconfirmed, and source-level scores vary far more than 0.5 points.

6 Results

Unless stated otherwise, scores come from the final models at 16-bit precision. The products available today are 4-bit builds, whose results we report separately. The breadth report and When2Call were run against deployed 4-bit endpoints.

6.1 English public suites

Table 3 gives results on the public suites, with Jev and Kev figures as reported by the Kev project [2]. Jev has only been run on development splits, so test and development results cannot be compared across systems. On the Transfer v4 development split, Zuse-27B is 2.5 points below Jev. An unpaired test on that accuracy gap gives $p = 0.19$, but paired predictions are not available, so this is only indicative. Jev is also ahead on Brier score and auto@5. Kev-27B is 2.9 points ahead of Zuse-27B on the test split.

6.2 Breadth and tool choice

Zuse-27B is behind Jev on the breadth report, and not clearly separated from Kev-27B. Table 4 shows a chance-corrected index from 0 to 100, built from 14 public datasets. The difference from Jev is 5.8 points (95% paired bootstrap CI 2.2 to 9.0). The difference from Kev-27B is 2.7 points (95% CI −0.2 to 5.3).

The Jev and Kev rows were measured by the Kev project [2]. The Jev rows were collected in September

Table 2: Evaluation suites and their use during development.

Suite	Content	Size	Use during development
European suite	11 sources excluded from training, 9 task groups, 27 languages	20,044 rows	Model selection, error diagnosis, design of later runs
Transfer v4 and v9 [2]	Public English suites	764 rows in Transfer v4 per split	Development split used for selection and data choices; test split scored on five 27B checkpoints
SemIf [2]	Authored English decisions plus perturbations	144 and 108 rows	Monitored during development
Scienthoon v1 [2]	English science reading	873 questions	Scored after freezing
Breadth report [2]	14 public datasets in 5 areas	1,990 records, development split	Scored after freezing
When2Call [20]	Tool-use choice	3,423 tasks	Scored after freezing
World languages	Belebele [11] and SIB-200 [12]	210 languages, 32,700 rows	Scored after freezing; European rows overlap the European suite's sources
Long documents	Synthetic documents, 2k to 32k tokens	1,350 rows	Scored on synthetic documents outside training

Table 3: English public suites. Cells show accuracy / Brier / auto@5 where available. Jev and Kev figures are as reported by Kev [2].

Suite and split	Zuse-27B	Zuse-9B	Zuse-4B	Jev	Kev-27B
Transfer v4, development	0.832 / 0.246 / 0.576	0.772 / 0.316 / 0.410	0.755 / 0.342 / 0.380	0.857 / 0.211 / 0.70	0.851 / 0.225 / –
Transfer v4, test	0.860 / 0.204 / 0.776	0.797 / 0.279 / 0.598	0.775 / 0.313 / 0.512	–	0.889 / 0.156 / –
Transfer v9, development	0.786 / 0.305 / 0.462	0.716 / 0.382 / 0.302	0.695 / 0.410 / 0.305	–	–
Transfer v9, test	0.787 / 0.285 / 0.609	0.723 / 0.369 / 0.485	0.694 / 0.415 / 0.392	–	–
SemIf, 144 authored rows	0.924	0.847	0.882	0.965	–
Scienthoon v1, development	0.767 / 0.308 / 0.585	0.679 / 0.392 / 0.309	0.685 / 0.388 / 0.435	–	–

Table 4: Breadth report on the development split, by area.

Area	Jev	Kev-27B	Z-27B	Z-9B
Knowledge, reasoning	37.7	33.8	36.0	24.6
Language understanding	77.4	75.6	69.7	57.8
Retrieval, classification	75.4	62.9	63.0	57.7
Tools, automation	63.6	64.2	59.0	58.5
Arts, human taste	12.7	14.7	10.0	6.7
Overall	**53.3**	**50.2**	**47.5**	**41.1**
95% CI lower	50.7	47.4	44.8	38.4
95% CI upper	56.3	53.3	50.7	44.2

2026 through a gateway, and the model revision is not exposed. Zuse rows are our own, and were run against a deployed 4-bit endpoint rather than the 16-bit evaluation build. That run met server errors, and all 1,990 records were answered after retries. This is a report in the style of a community index, not a third-party score.

On When2Call [20], a four-way choice (chance 25%) between using a tool, answering directly, asking for more information and saying it cannot help, and using our own instruction wording, Zuse-27B scored 57.5% and Zuse-9B 39.7% over 3,423 tasks. Mean confidence was 72.9% and 67.8%, so both models were overconfident on this task.

6.3 European suite

Table 5 gives the results on the European suite. The pooled calibration error is low, but this masks differences between groups. Within sources it ranges from 0.007 to 0.091, reaching 0.091 on the Norwegian quiz, 0.082 on grammar and 0.079 on multilingual MMLU. By language it reaches 0.067. Across the 14 breadth datasets, on the deployed endpoint, the mean is 0.100. Weighted by rows, the per-source error is 0.035, more than twice the pooled figure, because overconfidence on some sources offsets underconfidence on others. Calibration should therefore be checked for any specific task.

Figure 2 shows the automatable share by task group. It mostly follows accuracy (Spearman 0.93 across the nine groups), but accuracy is not corrected for chance. Nordic grammar is a clear case: 72.6% on balanced labels, with only 8% automatable at a 5% error budget. On Swiss court rulings Zuse-27B reaches 78.7%, below the 79.3% from always answering no, so it shows no gain over the majority answer, and only 1.5% can be automated. Banking intents and reading comprehension reach far higher shares. These are in-sample upper bounds with per-group thresholds (Section 5.2).

Figure 3 shows results by language. Icelandic (600 rows) and Bokmål (1,400 rows) sit within sampling

Table 5: European suite. Order flips use 1,223 choice rows and planted instructions a 1,500-row sample.

Measure	Zuse-27B	Zuse-9B	Zuse-4B
Accuracy (%)	**87.5**	84.1	81.2
Brier score	0.185	0.229	0.264
Expected calibration error	0.016	0.023	0.032
Automatable at 1 / 2 / 5 / 10% error (%)	40 / 58 / 77 / 94	30 / 50 / 67 / 86	26 / 42 / 61 / 78
Order flips, reversed order (%)	5.6	6.2	8.9
Order flips against a third order, single / two averaged (%)	3.7 / 3.3	4.4 / 3.5	6.9 / 5.2
Planted instruction followed, baseline in brackets (%)	8.1 (5.8)	12.2 (7.1)	14.2 (8.3)
Accuracy at 4-bit (%)	87.4	83.6	80.6
Brier score at 4-bit	0.188	0.236	0.275
Expected calibration error at 4-bit	0.016	0.026	0.034
auto@5 at 4-bit (%)	76.0	66.3	58.6
Separate long-document set, 2k to 32k tokens (%)	98.1	98.1	97.6

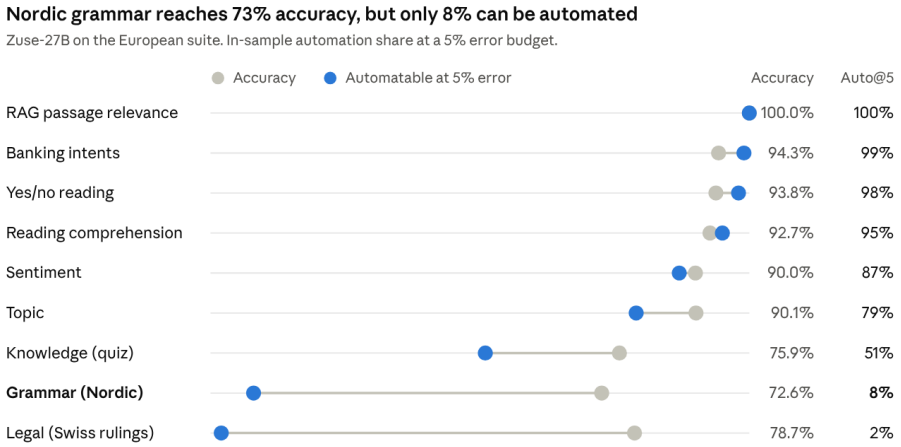


Figure 2: Accuracy and in-sample automation share by task group, Zuse-27B.

noise of 80%. Language averages reflect task mix as well as language, since Nordic languages carry the grammar and quiz sources. On Belebele reading and SIB-200 topics in the final model, those languages score between 87 and 96.

6.4 World languages

Table 6 reports reach across languages, including the languages trained on. Reading covers 122 languages from Belebele, and topic classification 205 from SIB-200, for 210 languages and 32,700 rows in total. We grouped languages using an approximate list of 140 codes derived by hand from the language list Qwen published for Qwen3, applied here to later Qwen versions.

On reading and topic together, Zuse-27B reaches 85% or above in 115 languages and falls below 60% in 25. The five weakest are languages in scripts the base model rarely saw, such as N'Ko, Ol Chiki and Tifinagh, at 22 to 31%. These have topic rows only, where uniform chance is 14.3% and always choosing the largest class gives 22 to 26%.

6.5 Inference speed

Table 7 reports model time at 16-bit precision on one B300 GPU, using Kev's serving stack. On 100 records, this path differed from the unoptimised path by at most 0.019 in probability, with at most one changed answer per model. In a 4-bit deployment on a 24 GB card, a two-question request took about 230 ms on a cached document. On 24 GB cards, documents are limited to 8,192 tokens.

7 Analysis of Training Choices

This section compares single-seed runs, so it is observational rather than a controlled study. Runs in a comparison may also differ in settings we do not report, so these comparisons are indicative. We treat gaps under 0.5 points on the European suite as noise (Section 5.3).

7.1 Base model and size

The largest effect we observed came from the base model. On an early corpus and identical data, moving from Qwen3.5-4B to Qwen3.8-27B raised accuracy on held-out sources from 75.4 to 85.4. On the same

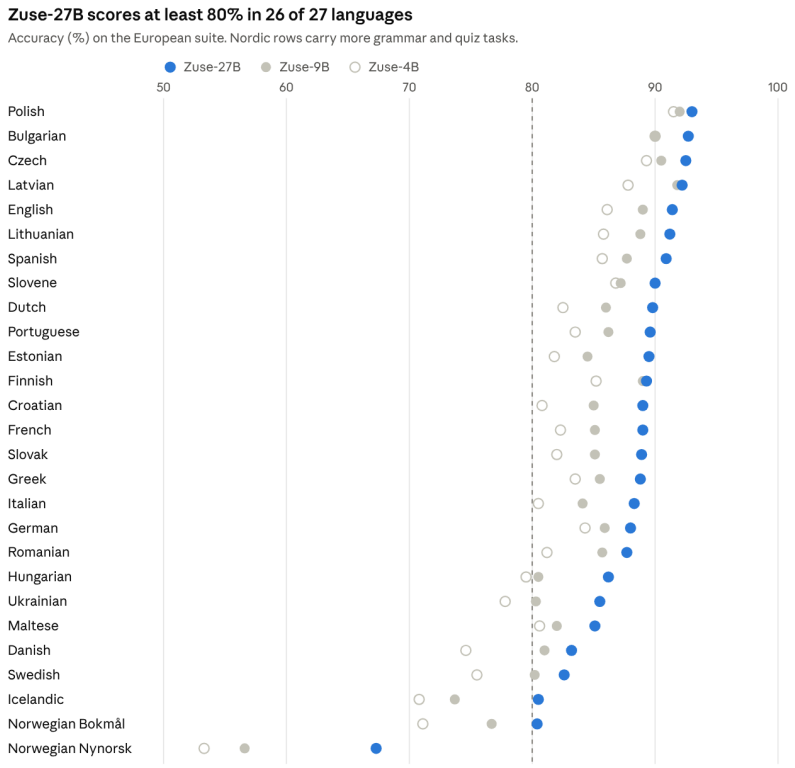


Figure 3: Accuracy by language on the European suite, all three models.

Table 6: Accuracy (%) on world languages, by whether the language is in the Qwen list. Cells show reading / topic.

Group	Reading languages	Topic languages	Zuse-27B	Zuse-9B	Zuse-4B
In the list	97	139	85.1 / 87.8	76.5 / 86.1	71.0 / 84.6
Outside the list	25	66	60.5 / 65.2	48.7 / 59.6	44.8 / 56.2

data and recipe, Qwen3.8-27B scored 85.6 against 75.8 for EuroLLM-22B [18], and led in all 27 languages. Borealis 2 31B [19], a Norwegian-trained model on a Gemma base, reached 82.9. These comparisons cannot separate parameter count from base-model quality.

World-language scores varied little across nine checkpoints of the three sizes. They stayed within 1.2 points of the final model for languages in the Qwen list and within 2.1 outside it. This suggests that reach outside the training languages comes largely from the base. The evidence is limited, because all nine checkpoints trained on similar data and there is no control.

7.2 Label-word collision

Adding one topic-labelled data family lowered accuracy on the SIB-200 topic test by about 10 points. The closest isolated pair of runs differs only in that family: 87.6 without it and 77.3 with it. Overall accuracy on the European suite moved from 86.7 to 83.4.

The cause appears to be shared label words. The added family used “geography” and “politics” with narrower meanings than everyday topic labels. On the affected run, the model chose “geography” 1,041 times where the correct count was 436. Datasets merged for training may therefore conflict through their label vocabulary even when their languages and domains look compatible.

7.3 Injection training and shortcut learning

Our first injection augmentation produced results that looked strong and proved misleading. On a Qwen3.5-4B run, following of planted instructions fell from 37.6% to 0.2%, but accuracy under attack rose above clean accuracy, from 75.6 to 81.7. Every planted note had named a wrong option, so the model could learn that a named option is incorrect.

We changed the augmentation so that planted notes sometimes name the correct option. The headline figure then looked worse, at 16.8% following against a 9.6% baseline, with accuracy under attack at 71.9 against 75.5 clean. This is the more honest result, because the model is no longer exploiting the construction of the attack. In the final Zuse-27B, following is 8.1% against a 5.8% baseline, and accuracy falls from 87.1 to 85.6 on the same sample. Robustness evaluations should therefore check for accuracy that

Table 7: Model time per request, new document / cached document, and throughput.

Model	2 questions (ms)	6 questions (ms)	5 questions, 2,200 tokens (ms)	Requests per second, 1 / 32 clients
Zuse-27B	32 / 18	45 / 31	182 / 51	20 / 47
Zuse-9B	13 / 8	17 / 12	86 / 20	49 / 118
Zuse-4B	10 / 6.5	14 / 10	84 / 18	62 / 141

improves under attack.

7.4 Order stability

Adding reordered copies of rows had almost no effect on order flips (13.6% to 13.1%, in a step that also added injection training), whereas a permutation-consistency loss helped. Across 27B runs, flips against a third order averaged 6.4% in twelve runs without the loss and 4.5% in eight runs with it. Those groups also differ in data, so the comparison is indicative. At a stronger setting on the 4B model, the loss cut flips from 10.5% to 6.8% and cost about 1.5 accuracy points. At a weaker setting on the 27B model we saw no accuracy cost.

8 Limitations and Threats to Validity

Model limitations. Knowledge-heavy questions remain weak in this training recipe: 75.9% on the quiz group, and 0.675 on MMLU-Pro [21] (200 development rows), equal to Kev-27B's reported 0.675 and below Jev's reported 0.84. Nynorsk scores 67.3%. Order sensitivity and instruction following are reduced but not removed. Accepted input runs to 64k tokens on larger GPUs, but accuracy above 32k is unmeasured because we lacked enough held-out documents. There is no image input.

Selection bias. The European suite shaped model selection, error diagnosis and the design of later runs, including removal of one data family and one targeted run. The Transfer v4 development split shaped data choices and flagship selection. Scores on both are likely to be optimistic. The English test splits were seen several times, and their scores were recorded alongside flagship decisions.

Missing comparisons. We have not run Jev, Kev or Clef on the European suite, so we make no claim about relative European performance. Nor have we compared frontier language models with structured output on the same rows.

Comparability. Jev and Kev figures come from the Kev project. Jev's revision is unknown and its automation figure uses a development split. Zuse's breadth scores came from a deployed endpoint rather than the evaluation build used elsewhere.

Statistical power. Results come from single seeds, and our noise estimate rests on two small-model retraining pairs. English test splits are small. Rows within the European suite are clustered by source, so simple margins understate uncertainty.

Construct validity. Not all training labels were independently checked. Long-document rows are simple fact checks. The planted-instruction test uses few wordings and always names a wrong option. All results come from public benchmarks or synthetic documents, not from a live deployment.

Undisclosed training details. We do not disclose the training data or the full training recipe, so the training cannot be reproduced from this paper, and readers must rely on our statement that the evaluation sources in the European suite were excluded from training. We compared the source identifiers in the final training list with the suite's 11 sources and found no match, but did not compare row text for that final list. Public suites can be run against the 4-bit Zuse-27B and Zuse-9B. Zuse-4B, the 16-bit builds and the sampled European suite rows are not available.

9 Conclusion

We presented Zuse, a family of adapter-based decision models trained for 27 European languages, and reported its results across English, European and world-language evaluations. On European sources excluded from training the largest model reaches 87.5% accuracy, though that figure is a development result. On English public suites it is behind Jev on most measures. Three lessons are likely to carry beyond this system: base-model choice had the largest effect among the comparisons reported here, overlapping label vocabulary between merged datasets appeared in one case to lower accuracy sharply, and robustness results should be checked for shortcuts. Accuracy also needs reading against chance level: the automatable share mostly followed accuracy, and the low shares came on tasks where accuracy sat near a simple baseline.

The priority for future work is a clean evaluation: a new European test suite, scored once and alongside open decision models. Repeated seeds for the main comparisons would resolve several of the confounds identified here.

Availability

Readers can test Zuse-27B through a live demo at `https://bineric.com/zuse`, which needs no API key, or through the Bineric API. Zuse-27B and Zuse-9B can also be run in a reader's own cloud account as 4-bit models through AWS Marketplace, and an Azure Marketplace offer is in preparation. Deployment guides and sample inputs are at `https://github.com/binericai/models/tree/main/zuse`. Zuse weights are not openly released, and Zuse-4B is not offered. The training data and training code are not released.

Competing Interests

The author works at Bineric AS, which sells Zuse. AI tools assisted with drafting and auditing this paper.

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